

increased, the weights of correctly classified data points are decreased, and this data is fed to the next classifier. This normalises the weights of all data points.

4. Steps 2 and 3 are repeated till you get the desired output.

Popular boosting techniques are adaptive boosting (ADA boost), gradient boosting, extreme gradient boosting (XG boost), etc.

Stacking

Stacking is a technique where heterogeneous learners are combined and fit on the complete data set, instead of a sample of a data set (as done in bagging). A single model is used to best combine the predictions from the contributing models instead of a sequence of models that correct the predictions of prior models (as done in boosting). In short, all the inner models are different from each other and fit on the same data set.

There are two components involved in stacking.

- a. *Level-0 models (base-models):*
These models fit on the training data and their predictions are compiled.
- b. *Level-1 models (meta-models):*
These are models that learn how to best combine the predictions of the base models and give the final prediction.

Voting

The voting technique is based on the concept of ‘wisdom of the crowd’. It takes a ‘vote’ of all the inner models’ predictions and returns the result with the majority of the votes as the final prediction. Voting can be of two types.

Hard voting: In hard voting, the outcome that has the most votes is selected as the final output. This technique has a drawback, as the final prediction of each inner model is taken at face value and tallied, without considering the confidence of the prediction and other relevant attributes.



Figure 3: Boosting

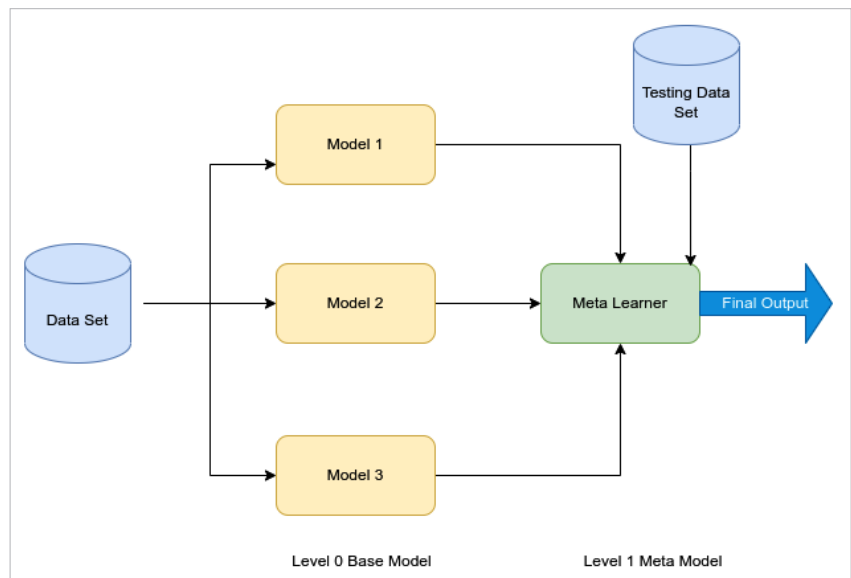


Figure 4: Stacking

Soft voting: In soft voting, the actual confidence probabilities of each prediction are taken into account. An average of all the confidence probabilities for each predicted label for every model is taken.

The fundamental difference between voting and stacking is how the final aggregation is done. In voting, user-specified weights are used to combine the classifiers, whereas stacking performs this aggregation by using a meta classifier. **END** 🐧

References

- <https://www.geeksforgeeks.org/xgboost/>
- <https://towardsdatascience.com/>

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